Proof of Mind

A Value Chain for Verified Mental Contribution

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noesis

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Abstract

Cryptoeconomics mints what a market needs to exist, and the market follows. Bitcoin is the proof: by letting strangers agree on who owns what without a trusted third party, it brought into being a market that could not exist before, the market for trustless ownership. This paper supplies the next missing medium, for the market in contribution: the chain itself becomes that market, pricing each contribution by the realized downstream work that builds on it, and makes that price the object consensus is about, not who holds the coin but how much each participant actually contributed. The claim is larger than a better proof of work, because measuring contribution on the chain itself unlocks two things a possession chain cannot reach.

The first is **cooperative economics by construction**. For a decade, decentralized systems located cooperation in a norm, a regulator, or a vote, each of them defectable, and each reproduced extraction by a new route. Here cooperation is moved into consensus itself: the layer where participants are alike is mutualized as a soulbound standing that secures the chain and cannot be bought, and the layer where they differ is a transferable, tradable state-capacity. You can buy storage; you cannot buy consensus. Cooperation becomes the Nash equilibrium, the strategy each participant prefers given what everyone else does, because the mechanism conserves value when you cooperate and fragments it when you do not, not because anyone promised to play fair.

The second is **convergence instead of competition**. The convergence of blockchain, AI, and attribution economics offers people a coordination interface they have not had before. Because value is measured on-chain and conserved along provenance, a contribution from any other chain can be imported, re-scored by the same value function, and credited on one canonical graph without loss. The chains that came before are not competitors to displace but the prototypes this design was learned from, and the question of which chain wins becomes all of them, on one attribution graph. To every chain that came before, the offer is not *come compete*; it is *come join us*.

The mechanism underneath both is *Proof of Mind* (PoM): value *created* as units of contribution, *measured* by a cooperative-game value (the Myerson value over a provenance graph), *owned and transferred* like a UTXO, and used to *secure consensus*, with verified contribution in place of expended energy. The one hard problem is measuring what a contribution is really worth when no market prices it. Any fixed formula is gamed the moment it is public, so the answer is not a fixed formula but augmented mechanism design: lock a few rules that can never be broken, and let the measure adapt and grow faster than anyone can game it. That room to grow is what makes the measure un-gameable, and it is the heart of the system. Any one of these properties is straightforward on its own. What this paper offers is the explanation of how they hold together at once.

These are claims with a structural argument behind them, offered open to refutation, not asserted. We mark throughout what is demonstrated and what is designed: the core mechanisms run and are tested in an open reference node; the cross-chain merge is a stated architecture, not yet a built feature; and feeding the value-model real outcomes at scale is the work that remains.

1 Introduction: the value chain Bitcoin is mistaken for

Proof of work was the first time a network agreed, with no one in charge, on a shared record of who owns what. Everything in this paper is built on that result. It secured *order*: a miner expends energy, the network agrees on a sequence, and the worth of the coin is settled elsewhere, by a market the chain never sees. What it does not record is *who created what value*, and for a chain about possession it did not need to. That is the next thing to build, not a flaw to fix: agreement over possession, then agreement over contribution.

A real value chain would record the contribution itself: each unit of value, measured by what it added, tied to whoever made it, transferable, and counted toward consensus. Chains like this are rare, and not because of ideology. They are hard to build. A possession chain never has to *measure* value; the market does that for it. A value chain has to. The hard part has a name: the gap between what a chain can check and what is actually true in the world, the *blockchain–reality airgap.* Bitcoin steps around it by letting a market price the coin. A value chain has to cross it, honestly and without being gamed. That is the problem this paper takes seriously. The same gap shows up everywhere in the design: between one mind and another (whom to trust), and between this chain and the chains it could absorb (Sections 4 and after).

Proof of Mind is the answer. It sits at the end of one line of descent: proof-of-work, then proof-of-*useful*-work (value measured by the signal a contribution carries, not the energy it burns), then **proof of mind**, a proof of verified, synergy-weighted contribution. The unit of work is a block of thought. The proof is that the thought added measurable value. Figure 1 shows how one unit of contribution becomes both consensus weight and tradable state.

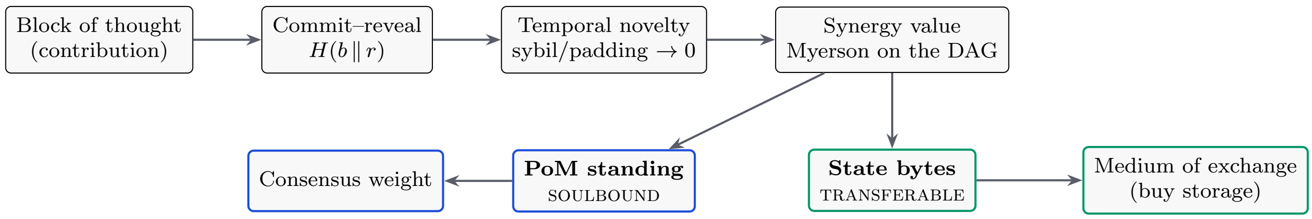

Figure 1: The contribution pipeline. A signed, provenance-complete block is ordered by commit–reveal, priced by temporal novelty, aggregated by a synergy value function, and split into a soulbound standing (consensus weight, cannot be sold) and a transferable state-capacity (a tradable commodity). You can buy storage; you cannot buy consensus.

2 Blocks and provenance

A block is the unit a participant produces, $b = \{\mathrm{id},\ \mathrm{parent},\ t,\ \mathrm{inputs},\ \mathrm{output},\ h\}$. Inputs are retained so that *how* the output came about is provable and reproducible: provenance, not merely possession. **Commit–reveal** makes authorship un-front-runnable. A producer publishes a commitment

$$c \;=\; H(b \,\|\, r) \tag{1}$$

with a signature and timestamp *before* revealing the content b and the secret r, binding authorship and ordering before disclosure, the same mechanism a fair batch exchange uses against the transaction-reordering attacks catalogued in Flash Boys 2.0. Committing and then failing to reveal valid content is slashable. The commit order is not only an anti-front-running device: it supplies the canonical ordering that makes the value rule of Section 4 strategyproof.

3 Ownership: Bitcoin-shaped, recursive for co-authorship

Each block is locked to an owner public key. Only the current owner's key produces a valid attestation, and ownership is transferable: the owner signs a reassignment to a new key, exactly as a UTXO is spent to a new locking script. Current ownership is not stored as a mutable table; it is *derived* by folding a signed transfer log over a genesis owner,

$$\mathrm{owner}_t \;=\; \mathrm{fold}\big(\mathrm{owner}_0,\ [\tau_1, \tau_2, \ldots, \tau_t]\big), \tag{2}$$

so there is no ownership record to forge; there is only a signature chain to verify. Transfer voids the prior attestation, and the new owner must re-sign.

A real block often has several authors: a prompt, a generation, a correction. Ownership is then a *share vector*, a set of (contributor, share) pairs, like a multi-output transaction. The within-block shares are computed by the *same value mechanism one level down*: contributors are players in an intra-block coalition game, and their shares are the Myerson value of their marginal contributions to that block's output. The economy is two-level recursive, outcome → blocks → contributors, and PoM sums a contributor's shares across all blocks.

4 Endogenous value: an honest, synergy-aware price

The chain is the market that contribution never had. A possession chain leaves value to an external market that prices its coin; this chain prices contribution *itself*, by the realized downstream flow a block generates along its provenance. The central object is that price: a value function $v(S)$ over sets of blocks, and the score, the stake, and the consensus weight are all computed from it. A market does not compute a true value from outside; it reveals one through what later work builds on, and that is what v does. The naive choices of v are trivial or gameable; this section gives one that is neither, and the rest of the paper is what it takes for that price to clear honestly.

One rule generates the rest. On a market that pays for contribution, the only thing worth paying for is *novel realized downstream flow along provenance*: value a block adds, that no earlier block already added, that later work actually builds on, credited back along the edges that produced it. The mechanisms in this section are not four defenses bolted on against four attacks; they are four faces of that one rule. Temporal novelty is its *no earlier block already added it* clause: you cannot be paid twice for the same coverage. Saturation is its *realized* clause: a finite outcome can be built on only finitely, so aggregated flow is bounded and volume cannot pump it (the particular decay used to allocate that bounded flow is a design choice, the one knob the rule does not fix). The HodgeRank residual is its *along provenance* clause made checkable: a collusive ring is circulation, not downstream flow, and the same computation that prices the work detects the difference in one pass. The bounded learned gate keeps the estimator faithful to the rule: the model may only estimate realized flow, never mint it. Read the subsections that follow as consequences of that one sentence, not as a list of separate parts.

4.1 Elicitation: from pairwise judgment to cardinal value

Human and model judgments arrive as *pairwise* preferences: block i contributed more than block j. Under the Bradley–Terry model each item has a latent strength s_i and

$$\Pr[\, i \succ j \,] \;=\; \frac{e^{s_i}}{e^{s_i} + e^{s_j}} \;=\; \sigma(s_i - s_j), \tag{3}$$

which is logistic regression on the score difference; the maximum-likelihood fit is convex and unique whenever the comparison graph is strongly connected. Generalized from a fixed set of items to a learned function $r_\theta(x)$ of block features, the same objective

$$\mathcal{L}(\theta) \;=\; -\sum_{(i,j)\,:\,i\succ j} \log \sigma\big(r_\theta(x_i) - r_\theta(x_j)\big) \tag{4}$$

is exactly the loss used to train an RLHF reward model from human preferences (Christiano et al., 2017). So from many "A contributed more than B" judgments we fit one "how much it added" score per block, the way a chess rating or an RLHF reward model is learned from pairwise wins. The distilled judgment model and the reward model are the same object, and that object is the production evaluator for v.

4.2 The additivity trap

A natural first attempt sets a block's value to its share of pairwise wins. We built it, and it failed in a useful way. If the coalition value is additive, $v(S) = \sum_{i\in S} v(\{i\})$, then the Shapley value collapses to the standalone value, $\phi_i = v(\{i\})$, and the normalized allocation is just the proportional (Copeland) win-share.

Claim 1. *For an additive game the entire $2^{|N|}$-coalition Shapley computation returns the same numbers as a one-line normalization. The cooperative machinery does no work.*

Pairwise comparison gives a *ranking*, which is additive by construction, so running Shapley over it does nothing. Measuring value matters only when $v(S)$ has **synergy**: a group is worth more, or less, than its members added up alone, because parts complete each other, repeat each other, or turn out to be pivotal.

4.3 Synergy and the Myerson value

We therefore define $v(S)$ as an *outcome value*: the quality or completeness of the outcome reconstructable using only the blocks in S. A block the outcome fails without is pivotal and earns a high marginal value; a block redundant with others earns little. Credit is the **Myerson value**, the Shapley value of the graph-restricted game in which only coalitions connected in the provenance graph G create value:

$$\phi_i(v) \;=\; \sum_{S\subseteq N\setminus\{i\}} \frac{|S|!\,(|N|-|S|-1)!}{|N|!}\,\big[\,v^G(S\cup\{i\}) - v^G(S)\,\big], \qquad v^G(S) = \sum_{T\in S/G} v(T), \tag{5}$$

where S/G are the connected pieces of S in G. So a block's credit is its average added value over every order the blocks could have been put together in, counting only the groups that provenance actually wired together. Value flows along provenance edges, not over arbitrary sets; plain Shapley is just the special case where everything connects to everything. Computing it exactly is exponential, so we estimate ϕ_i by random-order sampling (Data-Shapley). In a working prototype over real blocks, switching from the additive game to a coverage-based outcome value moved the Myerson credits well away from the proportional baseline, by a mean absolute difference $\|\phi - \text{Copeland}\|_1 \approx 0.26$: the cooperative value started doing real work, rewarding pivotal blocks and discounting redundant ones.

4.4 Strategyproofness: temporal novelty

A value function over content is gameable by repetition unless novelty is priced. The commit order does exactly that. A block's value is its coverage *minus* the coverage already claimed by all earlier-committed blocks,

$$\nu(b) \;=\; \Big|\; \mathrm{cov}(b) \;\setminus\; \bigcup_{c\,:\,t_c<t_b} \mathrm{cov}(c) \;\Big|. \tag{6}$$

You are paid only for what you add that no earlier block already covered. First to *commit*, then validly reveal, earns the novelty; later padding or sybil copies earn zero (the commit timestamp fixes priority, and the reveal merely claims it). (This is the second reason the commit order of Section 2 is load-bearing: it supplies the canonical order that makes the rule strategyproof.) In the reference implementation a sybil ring of five identical-content identities banks the coverage of at most one block (1.06×, not 5×), and a cross-block re-post earns zero.

4.5 Saturation under coordinated volume

Novelty defeats duplication, but a subtler attack remains: a vested participant, or several colluding ones, can post many distinct-but-low-value children of a valuable parent and try to pump the parent's aggregated flow. We damp this with a single geometric decay over the global, canonically-ordered list of a parent's contributing children. With decay $\rho = 1/\varphi$ (the inverse golden ratio), the parent's aggregated contribution is

$$F \;=\; \sum_{j\geq 0} \rho^{\,j}\, c_{(j)} \;\leq\; \frac{c_{\max}}{1-\rho} \;=\; \varphi^2\, c_{\max} \;\approx\; 2.618\, c_{\max}, \tag{7}$$

a convergent series: coordinated volume *saturates* rather than amplifies. In plain terms, each extra post counts for geometrically less, so flooding the system hits a fixed ceiling instead of compounding, and the total can never run away no matter how many copies are posted. An earlier design that damped two axes separately (within an identity and across identities) left their product unbounded, a "diagonal" pump in which K coordinated identities each posting M children compounded both tails. Folding both into one joint decay over the flattened global order closes it. Figure 2 shows the measured effect: before the fix the diagonal grows past the honest single-identity ceiling; after it, the whole grid is bounded by that ceiling.

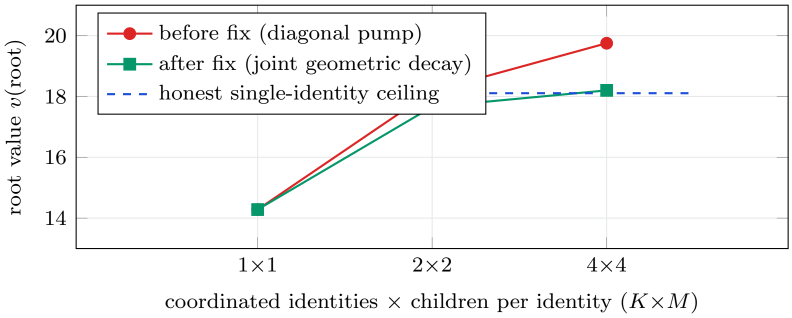

Figure 2: Coordinated-volume saturation, measured in the reference node. Before the joint-decay fix, K coordinated identities each posting M valueless children pump the root's aggregated value past the honest single-identity ceiling (red). After folding the two damping axes into one geometric decay over the global canonical order (green), the entire $K\times M$ grid is bounded by that ceiling to within 2%. Honest single contributions are paid unchanged.

4.6 A worked example

A small example ties the pieces together. Take a result A with coverage $\{1, 2, 3\}$ committed first, a block B built on A with coverage $\{3, 4\}$ committed second, and a near-duplicate D of B committed third. Temporal novelty credits coverage only against everything committed earlier: $\nu(A) = |\{1, 2, 3\}| = 3$, $\nu(B) = |\{4\}| = 1$ (shingle 3 was already claimed by A), and $\nu(D) = 0$ (its coverage is wholly claimed). Padding and sybil copies earn nothing, concretely.

For credit, let the outcome value of a connected set be its covered shingles, $v(S) = \left|\bigcup_{i\in S} \mathrm{cov}(i)\right|$, so $v(\{A\}) = 3$, $v(\{B\}) = 2$, and $v(\{A, B\}) = |\{1, 2, 3, 4\}| = 4$. The Myerson value over the edge A–B is $\phi_A = \frac{1}{2}(3-0)+\frac{1}{2}(4-2) = 2.5$ and $\phi_B = \frac{1}{2}(2-0) + \frac{1}{2}(4-3) = 1.5$, summing to $v(\{A, B\}) = 4$. A proportional split by standalone value would instead give $A = 2.4$, $B = 1.6$: the synergy rule shifts 0.1 of credit *to* the pivotal A (which owns the shared shingle 3) and away from the partly-redundant B. The cooperative machinery is doing work, not ceremony. Coverage is submodular, so on its own it can only *discount redundancy*; crediting genuine complementarity — work worth more together than apart — is what the learned $v(S)$ adds.

Finally, suppose an attacker floods A with N low-value children each contributing c. The geometric decay above bounds the parent's aggregated flow at $\sum_{j\ge 0} \rho^j c = \varphi^2 c \approx 2.618\, c$ for *any* N: volume cannot pump the gate.

4.7 Certifying the value: the HodgeRank residual

Pairwise judgments need not line up consistently; the contradictions are built in (Condorcet), not just noise. A single value number is the right object only when the judgments nearly agree. We make that checkable. Writing the comparisons as a flow Y on the *comparison graph* (the graph of judgments, distinct from the provenance graph that restricts the Myerson value) and solving the weighted least-squares problem

$$\min_s \sum_{i,j} w_{ij} \left(s_i - s_j - Y_{ij}\right)^2 \tag{8}$$

yields a single ranking s (a Laplacian solve, a generalization of Borda counting). The Helmholtz–Hodge decomposition then splits that flow into three independent parts,

$$Y = \underbrace{\mathrm{grad}\, s}_{\text{global ranking}} \oplus \underbrace{\ker \Delta_1}_{\text{harmonic (global cycles)}} \oplus \underbrace{\mathrm{curl}^*}_{\text{local cycles}} . \tag{9}$$

Plainly: the same computation that ranks the blocks also measures how self-consistent the underlying judgments were and points at exactly where they contradict themselves. The residual energy is not an error bar; it is a *certificate*. A large curl component says fine-scale ordering is unreliable; a large harmonic component says the judgments cycle through holes in the graph. Where the Shapley/Myerson construction *assumes* a consistent scalar world (its axioms force the residual to zero), HodgeRank *measures* whether that world holds, and localizes where it breaks. This matters for security as well as honesty: a coordinated sybil or collusive ring injects detectable harmonic circulation, so a spike in harmonic energy is a manipulation alarm in the same computation that produces the value. The value layer therefore ships its own certificate of when the single number it reports can be trusted. A large residual is not merely reported: it routes the contested credit away from the scalar to a higher-cost path, a multi-dimensional split or a human jury, spent only where the judgments actually disagree.

4.8 The learned evaluator, bounded by construction

The production $v(S)$ is a learned reward model, and learned models drift and can be probed. We do not ask it to be un-gameable. We deny it the authority to be dangerous. The part that actually protects the system is the realized gate

$$\text{seed}_i \;=\; \bar{\nu}_i \cdot g(f_i) \cdot \omega(\ell_i), \qquad \bar{\nu}_i,\; g(f_i),\; \omega(\ell_i) \in [0,1], \tag{10}$$

the product of floored novelty $\bar{\nu}_i$, a realized-flow gate $g(f_i)$, and the learned outcome ω scored on the block's provenance lineage ℓ_i. The factors are multiplicative and bounded in $[0, 1]$, a chain of dials each set between 0 and 1, so every one of them (the learned factor included) can only turn the payout *down*, never conjure it up. The learned model is confined to two roles that have no minting power: it may offer an honest contributor an early liquidity *advance* against expected vesting (its own capital at risk, repaid from realized value and slashed on shortfall), and it may enter a dispute as *evidence* the jurors weigh. A corrupt model that scores everything maximally reduces to the un-learned gate exactly: in the reference node, $v_8 \equiv v_7$ under an adversarial $+\infty$ model. The proof obligation collapses from "the neural network is un-gameable" to "three bounds hold," which are small functions with tests.

The value of the learned model is what it adds *within* those bounds. Figure 3 reports a held-out measurement: trained on one split of coalitions and tested on coalitions never seen, the learned $v(S)$ ranks "work built on work" above the same cells dumped as orphans with accuracy ≥ 0.9, while the coverage proxy is blind to lineage and ties at chance.

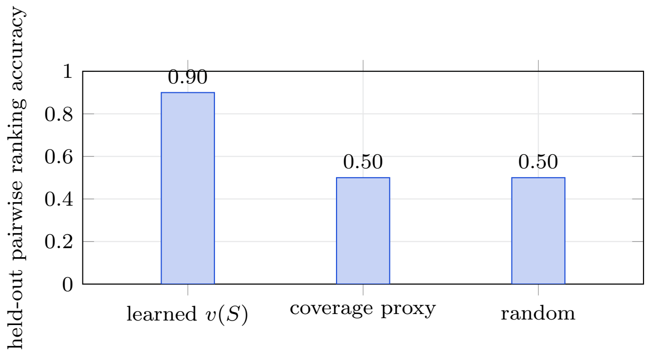

Figure 3: The moat, measured not asserted. On coalitions never seen in training, the learned $v(S)$ distinguishes connected, synergistic contribution from the same content as disconnected orphans (≥ 0.9), while the coverage proxy the per-block rule can see is blind to lineage and ranks at chance (0.5). The remaining mile is real outcome-preference data; the measurement harness runs unchanged when it lands.

5 Measurement as a living mechanism

The value function of Section 4 is not a fixed formula, and it cannot be one. Any static value rule is Goodhart-bait: the moment it is published, it is gamed, because a fixed objective tells an adversary exactly what to optimize. This is reward-model overoptimization (Gao et al., 2023) made adversarial, and more generally the performativity of any published predictor (Perdomo et al., 2020): the act of publishing the measure changes the distribution it is measured on. The only un-gameable measure is one that adapts to the gaming. Measurement on this chain is therefore a closed loop, not a closed form, and its components, introduced separately above, are one adaptive system.

1. **Realized, not predicted.** The learned model can only lower a seed it never raises one (Section 4). Value is gated on realized downstream flow, so the measure is anchored to what actually happened, not to a prediction an adversary can spoof.
2. **It retrains on outcomes.** The chain is a clean, provenance-complete, value-weighted dataset (Section 9). The evaluator is fine-tuned on verified high-value contributions against caught failures, so the measure sharpens as the chain accumulates verified truth. Yesterday's attack becomes today's training signal.
3. **Value is provisional, then adjudicated.** A payout is not final at emission. A challenge window lets a bonded challenger refute a value claim; a refuted claim is zeroed and the claimant slashed. Measurement is a dispute process with finality, not a one-shot score.
4. **It cannot be banked.** Standing decays, so a high score earned once does not persist without continued contribution. The measure tracks live value, and a stale position is reclaimed.
5. **It diagnoses its own breakdown.** The HodgeRank residual (Section 4) is a continuous detector: when the judgments stop fitting a single scalar, through genuine multidimensionality or through a coordinated manipulation injecting harmonic circulation, the residual energy rises and flags it. The system therefore knows

when to distrust its own number, rather than reporting a confident scalar over a structure that no longer supports one.

6. **The dimensions of value can themselves evolve.** The value matrix is governance-mutable but verifier-gated: a new dimension is admitted only if it predicts realized downstream value, weights move within bounded ranges, and no real dimension can be zeroed. The measure is not frozen at genesis, and it cannot be captured into a private one.

Put together, the loop is: measure, realize, dispute, retrain, decay, re-measure. A static rule loses to Goodhart on its first day; an adaptive measure with an un-gameable realized-value anchor, a dispute process for finality, and a residual that reports its own validity does not present a fixed target to optimize against. This is why Section 10 calls value measurement the load-bearing open problem rather than a solved one: the claim is not a closed-form proof of un-gameability but a mechanism that holds un-gameability as a moving equilibrium, and whose present strength is measured (Figure 3) rather than asserted. A static rule, exact on paper but gameable in practice, is wrong in the only sense that matters; an adaptive measure that is approximately right and provably converging is the better object. It is better to be approximately right than absolutely wrong.

6 Proof of Mind

A participant's **PoM score** is its accumulated Myerson value over the verified, owned, provenance-complete blocks it holds,

$$\mathrm{PoM}(a) \;=\; \sum_{b\,\in\,\mathcal{B}(a)} \phi_b, \qquad \mathcal{B}(a) = \{\text{verified, owned, provenance-complete blocks of } a\}. \tag{11}$$

In plain terms, your standing is just the summed, verified value of the work you own, nothing you can buy, only contribution that survived checking. Three properties are inherited from the construction. It is **verifiable**: every contributing block is signed and provenance-complete, so anyone can recompute the score from public data, or, where the content is held private, verify a succinct proof that the score was computed correctly (Section 11). It is **sybil-resistant structurally**: PoM requires owned blocks whose value was synergy-judged, and splitting one mind across many accounts does not multiply PoM because the synergy game discounts redundant copies; a thousand empty nodes score zero. Proof-of-work buys this same sybil and spam resistance with a cost *external* to the chain (expended energy); PoM derives it *endogenously* from the value rule itself, so spamming and splitting earn nothing without any energy price to pay. And it is **earned, not bought**: the stake is accumulated proof of contribution, and slashing revokes PoM for refuted blocks, caught hallucinations, and failed attestations.

6.1 Theory of Mind to Proof of Mind

Across agents, Theory of Mind is inference. You cannot see inside another mind, so you guess whether to trust it. That guess is an airgap, the inter-mind version of the gap between a blockchain and reality. Economic Theory of Mind reframes a mind as an economy whose outputs carry value. Proof of Mind is the verifiable economic proof of that economy, and it closes the airgap: instead of each node inferring whether another is a mind worth trusting (intractable and game-able), PoM supplies a proof anyone can recompute. Trust stops being a per-mind inference and becomes a structural property. The chain is capacity → mind-as-economy → proof of that economy.

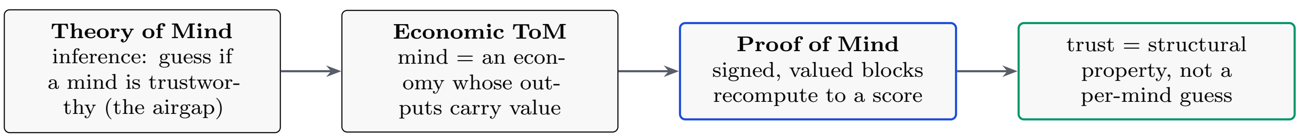

Figure 4: The mapping that makes a decentralized network of minds tractable: replace "infer whether to trust" with "verify a proof."

7 Consensus

Validators are weighted by **PoM**, not by energy or capital. The tamper-evident, signed, owned chain is the ledger; PoM is the stake; consensus is PoM-weighted agreement on the canonical chain. Three design choices make this safe.

7.1 Retention decay and the finalization basis

Vote weight decays with staleness, so standing tracks live participation rather than past glory. Decay applies to the *franchise*, the vote weight, never to the staked balance, and symmetrically across all dimensions so the effective mix is time-invariant. The remaining question is what a supermajority is a fraction *of*. A base-weight denominator is eclipse-resistant but halts under low participation; a pure effective-weight denominator removes the halt but lets an attacker shrink the denominator and finalize a minority. The stable choice is the hybrid

$$\text{finalize} \iff \forall d: \quad W^d_{\text{for}} \;\geq\; \tfrac{2}{3}\,\max\!\big(W^d_{\text{eff}},\, Q^d\big), \tag{12}$$

which pins the denominator at a quorum floor Q^d, a fraction of base weight, so neither failure occurs. In plain terms: a block is final once a two-thirds supermajority backs it — the Byzantine-fault threshold of PBFT, carried into a finality gadget in the manner of Casper — measured against whichever is larger, who is actually voting or a fixed floor, so an attacker can neither stall the chain by sitting out nor finalize a minority by shrinking the pool. The composition over the dimensions d is AND-gated, not a substitutable weighted sum: a single dimension below the supermajority cannot finalize alone. This is verified against tested reference models, including a self-audit that demonstrates the eclipse on a bare effective basis and its closure under the quorum floor.

7.2 Stability: no profitable fork

PoM-weighted agreement must be defection-proof. Individual deviation is already deterred by the Nash structure of the value rule; the remaining concern is the *coalitional* one, that no group of validators should profit by deviating together. We impose a coalitional stability constraint over the PoM-weighted game: require the allocation to lie in the **core** when it is non-empty (no sub-coalition does better by forking) — the coalitional-stability lineage of Aumann's strong equilibrium and the coalition-proof equilibria of Bernheim, Peleg and Whinston — and fall back to the **nucleolus** when the core is empty (the unique split that leaves the most-tempted coalition with the least reason to defect). This is added precisely because consensus requires it; pure attribution does not need a stability concept.

7.3 Three powers: cognition leads, compute and capital keep it live

Consensus draws on three independent powers, **cognition** (PoM), **capital** (state-stake), and **compute** (proof-of-work, on a separate money layer), and they do two different jobs. Cognition is the *franchise*: PoM is earned by verified contribution, cannot be bought, and is what confers standing and leads the finalization of value. Compute and capital are the *liveness floor*: contribution is bursty, but block production cannot be, so when no minds are contributing, miners and stakers still produce and order blocks and the chain does not halt for want of new thought.

The division is strict, and it is what keeps the headline true. The floor may advance the chain and order pending transfers, but it earns no PoM, mints no standing, and cannot rewrite attribution or value-history. You can keep the lights on with compute or capital; you still cannot buy consensus weight with either. PoM leads and re-asserts as soon as contributors return; the floor is a fallback producer, never a value authority.

What "compute votes" means is narrow. The energy money the miner earns carries *zero* consensus weight: holding the coin buys no say. The small weight attached to proof-of-work belongs to the *act of mining*, the liveness-floor producer that orders blocks, and it is kept *out of finality* entirely. A miner's probabilistic lag is a finality-safety hazard, so finalizing a block is the job of capital and cognition (PoS and PoM) under an anti-concentration floor, never of the hash. Energy circulates; it does not vote on value.

Holding the powers separate is also what makes the chain capture-resistant. The rationale for exactly three is a non-dominated cycle: with two, one power tends to be the strict best response and captures the other; three admits a non-transitive (rock–paper–scissors) relation in which none dominates, and we judge further powers to add coordination cost without adding non-domination (multi-resource consensus in the manner of Minotaur composes compute and capital; we add the value franchise as the third). The asymmetry is deliberate, and it is what keeps "you cannot buy consensus" true even though bought capital shares the liveness floor: *halting* the chain requires defeating whichever power is awake at the floor, but *capturing value* requires defeating cognition specifically. Finality is the job of capital and cognition under an anti-concentration floor, never of the hash, and the value layer answers to cognition above all; so compute can keep the chain alive but cannot, even in coalition with capital, rewrite attribution or mint standing.

The compute layer is honest money, not waste. The floor's compute is a proof-of-work *money layer*, kept deliberately. Proof-of-work's energy cost is not a flaw peculiar to Bitcoin: it is the unavoidable price of issuing money with no privileged issuer, and every proposed alternative pays the same cost in a less visible form, an argument made forcefully by Sztorc (2015), under which locked stake, bought votes, and ground histories are all *obscured* proof-of-work. A chain that uses proof-of-work in its own consensus, as this one does, cannot coherently call it waste. What Noesis adds is usefulness by construction. The money layer is a proportional proof-of-work currency in the manner of Ergon: issuance scales with the work committed, so the coin is demand-elastic and anchored to the marginal cost of energy rather than to appreciation. A coin not expected to rise in value is one meant to be *spent*, and the proportional arms race makes this self-enforcing: miners must keep selling to cover ongoing power and hardware costs, so hoarding a non-appreciating coin is bad business and the coin circulates as a true medium of exchange. This is what proof-of-work does best, an *energy oracle* pricing real-world energy into money, without bolting a foreign task (primes, protein folds) onto the hash. Using proof-of-work here is therefore not a concession but a strength: cognition (PoM) secures value, energy prices money, and each tool sits where it fits.

Ergon alone, however, oscillates. The proportional reward couples price to hashrate, and a single feedback loop overshoots, so a proportional coin left to itself swings rather than settles. The money layer is therefore not Ergon as-is but a three-mechanism stabilizer we call the *Trinomial Stability System*, acting at three separated timescales on one energy-money token. The proportional proof-of-work reward is the long-term anchor, pegging value to energy cost. An elastic rebase corrects short-term price deviation. A PI controller damps the medium term. The result that makes this a system and not three knobs is that the three loops at separated frequencies are jointly stable where the proportional reward alone is not: the rebase and the controller are the damping that turns Ergon's oscillation into a stable, spendable unit of account. This layer is *designed, not yet built*; the reference stub's current issuance curve is a placeholder to be replaced with the proportional rule and its two damping loops.

The two layers are economic opposites by design, and that is why they fit. PoM is a *scarce, inelastic* standing that cannot be bought and carries the majority of consensus weight: earned value decides who governs. The energy money is *elastic* and meant to be spent: priced work is what circulates. Cognition governs and does not inflate; energy circulates and does not vote; capital is the stake of the stakeholders and their voice, the conventional role unchanged. Each does only what it is best at, and the whole holds because none is asked to do another's job.

8 Cryptoeconomics: one PoM, one byte

Storage is the scarce resource, and PoM standing is the right to occupy it: **one unit of PoM standing entitles its holder to mint one byte of transferable on-chain state**. Standing is the mint: it is issued, not bought, because novel verified contribution is what creates it. Two distinct sinks bound the supply, and keeping them distinct is what makes the accounting close. *Franchise decay* acts on the soulbound standing, so vote weight tracks live participation and a stale position is reclaimed; *state-rent* acts on the transferable bytes, paid by whoever currently holds the state, as in the cell-model rent of Nervos CKB. Supply is then bounded rather than balanced by assertion: total live standing converges to a fixed point at which the issuance rate from new contribution equals the decay rate integrated over the existing stock, and the decay constant is the single parameter that sets that ceiling. A floor protects a genuine contributor from being zeroed by a brief absence: decay reclaims idle capacity, it does not erase earned standing to nothing.

The subtlety that keeps this from collapsing into proof-of-stake is a split between two cells. A participant's **standing** is soulbound: it is the franchise (consensus weight and the right to mint state), and it cannot be sold. The **state-capacity** it earns the right to mint is transferable: a byte of state is a liquid commodity that trades freely. The result is that you can buy storage but not consensus. A transferable franchise would let a rich actor buy consensus, which is proof-of-stake by another name; a soulbound franchise makes "earned, not bought" and structural sybil-resistance real at once.

9 Backwards-enforcement of the model

The value chain is also a training signal. Each block is provenance-complete, owner-authenticated, and Myerson-valued: a clean, value-weighted dataset, exactly the object Data-Shapley attributes. With open model weights, fine-tuning on high-PoM verified blocks against caught hallucinations lets the governance layer's accumulated, verified truth shape the model. With closed weights, the same truth constrains the model in-context: gates block disallowed actions and the signed chain contradicts a hallucination, forcing correction. Governance produces a training signal, which shapes the model, which governance then verifies, and the loop compounds. The structure does not only constrain the mind; it teaches it.

10 Security and honest limitations

10.1 Threat model

The mechanisms above compose into one defense surface. Each attack is matched to the structural property that defeats it and its status in the reference node.

Attack	What defeats it (structural)	Status
Sybil split (one mind, many accounts)	synergy game discounts redundant copies; PoM does not multiply	demonstrated ($1.06\times$ at K=5)
Padding / re-post of coverage	temporal novelty: later copies earn 0 novel coverage	demonstrated
Coordinated-volume / diagonal pump	single joint geometric decay, bound $\varphi^2 c$ (Fig. 2)	demonstrated
Gamed learned evaluator	bounded role: can only lower a seed, never mint ($v_8 \equiv v_7$ under $+\infty$)	demonstrated
Cyclic / collusive judgments	Helmholtz–Hodge harmonic-energy alarm, with per-identity attribution feeding a slash	demonstrated (slash-wiring deferred)
Self-referential scoring	independent evaluators, eigenvector damping, synergy discount	designed
Eclipse (shrink the finality denominator)	hybrid max(effective, quorum-floor) basis	demonstrated (self-audit)
Profitable fork (coalition deviates)	core / nucleolus stability constraint	demonstrated (solver)
Spend another owner's cell	existence check + lock-signature control	designed (call-site wired, inert)
Theft by forking the network	attribution *is* the mechanism: stripping provenance is incoherent, keeping it credits the source	structural
Off-chain sale of a standing key	soulbound blocks on-chain transfer; the residual is selling the identity key itself, bounded because the key inherits the identity's full slashable history (and personhood-binding)	designed (residual)

The load-bearing bet. A value chain must measure value un-gameably across the blockchain–reality airgap (Section 4), the gap between what the chain can verify and what is true in the world. This is the problem Bitcoin sidesteps by being mere possession, letting an external market bridge the airgap for it. If the learned outcome evaluator is gameable, PoM degrades into a reputation system. This is the central open problem, and we treat it as such. The mitigations are structural: the realized-flow gate that the learned model can only lower, the temporal-novelty rule, the HodgeRank manipulation certificate, and the held-out measurement of Figure 3. The remaining mile is real outcome-preference data at scale.

Augmented mechanism design: the counterweight. The concession is bounded, not open-ended, and the bound is the methodology. We do not replace the cooperative-game mechanism with a learned oracle, nor do we trust the oracle; we *augment* the mechanism with invariants that hold by construction. The learned evaluator can only lower a seed, never mint one; value is gated on realized flow; supply is conserved; finality is an AND over independent dimensions. Each invariant is deterministic, so the blast radius of a wrong measurement is bounded *a priori*: at worst the chain under-credits honest work, it never forges value from a gamed score. Within those invariants the living-measurement loop (retrain on outcomes, dispute, decay, residual self-diagnosis) is a deterministic path that converges toward the true measurement as the chain accumulates verified outcomes. The open problem is therefore the distance between *invariant-bounded and converging* and *closed-form-proven*, not between working and not working. This is augmented mechanism design: augment a sound mechanism with math-enforced invariants rather than betting the system on a component being perfect, so the honest concession and a credible deterministic path coexist.

Self-reference. A block can carry information about contributions and thereby influence the scoring of others. The value layer scores such a block for its own marginal contribution; the elicitation layer consumes its content as a judgment about referenced blocks. These must stay decoupled, and no block may be the sole scorer of its own value. Guards: independent evaluators, eigenvector-style damping over the attribution graph, and the synergy game's discounting of self-referential coalitions.

Demonstrated versus designed. *Demonstrated* in the open reference node (a Rust implementation with a no-std verification core shared with on-chain scripts, 289 passing tests at the time of writing): ownership and transfer, per-block signing, tamper-resistance, the synergy value over a coverage proxy with Myerson and Bradley–Terry and permutation sampling, temporal novelty, coordinated-volume saturation (Figure 2), the orphan-root fan-out bounded by realized-flow gating, collusion detection on topology (the mutual-citation circulation alarm and its complement, the Helmholtz–Hodge harmonic-energy alarm that catches directed rings, with per-identity attribution that names the ring members and spares an honest contributor who builds on them once), the bounded learned evaluator and its held-out measurement (Figure 3), the dispute and slashing path, PoM-weighted finalization with the hybrid basis, the core/nucleolus stability solver, and the on-chain recomputation of the ordering and finalization rules inside the virtual machine. *Designed, not yet built*: the learned $v(S)$ on real outcome labels, the wiring of the collusion attribution into the dispute slash path, the production two-level recursion, owner-signature control of spends (the deterministic transaction digest is built and tested, and the per-input authorization gate is wired into ledger validation, inert until the signature verifier is linked at deploy), the open-weight fine-tune loop, and the money/stability layer.

11 Privacy: what a value chain can and cannot hide

A value chain is provenance-complete by necessity. Proof of Mind is verifiable precisely because anyone can recompute a score from public, signed, attributed blocks; that recomputability is the opposite of privacy. So Noesis cannot offer the secrecy of a possession chain that records only balances, it must expose enough for value to be checked. The honest question is not whether everything can be hidden (it cannot) but what minimum must be public for trustless verification, and how the rest is protected. The design separates three layers.

Identity is pseudonymous, not anonymous. Blocks lock to public keys, not legal identities, as on Bitcoin; standing accrues to a key. The tension is structural: sybil-resistance and the soulbound personhood-binding (Section 10) pull toward few keys per person, which limits anonymity-by-many-keys. Noesis therefore offers pseudonymity, not unlinkability: one mind to one accumulating standing is load-bearing for PoM, so a participant may be unknown but their contributions are linkable to the standing they earn.

Content can be committed without being published. The novelty layer already operates on *hashes*: coverage is a set of shingle commitments in the sparse-Merkle index (Section 4), not plaintext, so duplicate-detection and temporal novelty are computable without revealing the underlying text. A contributor can prove coverage and novelty against the public commitments while keeping the work itself private, in the spirit of a validity proof that checks a state transition without exposing its witness.

The hard case: value measurement requires observation. The learned evaluator $v(S)$ is the exception. Scoring a contribution's outcome-value requires *processing* its content, a privacy analog of the blockchain–reality airgap (Section 10): measuring value requires observing the thing measured. The logical resolution is *compute-to-data*, bring the evaluator's model to the content, never the content to the evaluator. The contribution stays in the contributor's environment; on the cell model the content cell is *referenced* read-only by the evaluating transaction and never consumed, so at the protocol level the data cannot leave the owner's control, and only the score and a proof leave to be committed on-chain. Three layers harden the result, in increasing cost: a *zero-knowledge proof of correct execution* binds the emitted score to the canonical model run on the committed content, removing any trust in whoever ran it; calibrated *differential-privacy* noise on the score bounds what the number itself can leak, since a crafted probe could otherwise reconstruct content through repeated scoring; and fully homomorphic encryption, where it becomes practical, is the frontier in which even the evaluating environment computes on ciphertext. Compute-to-data is the deployable sweet spot, strong privacy without the overhead of FHE or multi-party computation; ZK-execution and differential privacy close the two gaps it leaves (honest execution, output leakage). We mark this *designed, not built*: the reference node measures over public content, and the strategyproof skeleton (floored novelty, the realized-flow gate, the bounded learned factor) is unchanged whether the content is public or scored under compute-to-data, because each gate consumes a commitment, a proof, or a noised score, not the plaintext.

Data governance. Because the chain is also a training signal (Section 9), a participant's blocks shape the model. Consent is structural rather than promised: contribution is opt-in and attributed, the standing it earns is soulbound to a key the participant controls and can let decay, and the same proven-evaluation path that keeps content private from readers can keep it private from the training corpus while still admitting its verified value. Fully content-private contribution that nonetheless trains the model is the same observation tension above, and we treat it as open, not solved.

12 Fair launch and theft-resistance

The creator's pre-launch contribution would otherwise be an unearned head start. It is neutralized by a **genesis-burn**: the chain stays continuous from genesis and pre-launch blocks remain auditable, but their standing and state-value are programmatically burned to zero at the launch height. This is a fair launch one can verify on-chain, chosen over a chain reset, which only asserts fairness. Post-launch work may still build on a pre-launch block; the credit it generates flows forward to post-launch contributors along the provenance edges, while the pre-launch owner's burned standing is not retroactively restored.

The mechanism also yields an unusual theft-resistance. Because the system *is* attribution, a derivative deployment has two options. It can strip the provenance, in which case the attribution layer the system depends on is gone and the fork is incoherent; or it can preserve the provenance, in which case its edges trace back to this origin and running it credits the source. Copying the network honestly adds the copier to the same attribution graph, so there is no outside to fork to. Forking, which fragments value on a possession chain, here becomes contribution.

13 Forwards-compatibility: convergence, not competition

A possession chain fragments. Every new design competes to be the one chain, and value splits across the survivors. A value chain can do the opposite. Because contribution here is measured endogenously and conserved along provenance, a contribution from another chain can be imported as an attributed cell, re-scored by the same value function, and credited on one canonical graph without loss. The chain is not only backwards-compatible (continuous from genesis, prior blocks auditable); it is forwards-compatible, in that other useful-work chains can converge into it by a reverse fork. A normal fork diverges, splitting one chain's state into two; a reverse fork converges, merging many chains' contributions into one attribution graph.

This reframes the open question that hangs over every useful-work chain. The question "which one do we adopt" presumes a competition for a single standard. The answer is all of them. The useful work of Primecoin, of a storage market, of a decentralized training run, is in each case a contribution with provenance; a chain that ingests foreign contributions agnostically, scores them by one value function, and credits them on one canonical graph does not have to win the adoption fight, it absorbs it. This is how a fractured ecosystem un-fractures: not by forcing a million chains to adopt one of them, which is impossible, but by giving them a substrate they converge into without anyone losing what they built. The cooperation is structural, a property of conserved contribution, not a treaty that can be defected from. To every other chain, the offer is no longer *come compete*; it is *come join us*.

The enabling property is the endogenous measurement of Section 4: a possession chain has no on-chain quantity to carry another chain's value across a merge, a value chain does, so convergence is the ecosystem-scale expression of the same property that makes the single chain novel. It generalizes the theft-resistance of Section 12: where that argument left a fork no outside to escape *to*, this leaves a rival chain no outside to converge *from*, since its contributions either join this graph or strip the provenance that gave them worth.

This is a design direction, not a built feature, and it is honest about its hard parts. A chain-agnostic contribution adapter must map a foreign unit to a provenance-bearing cell; cross-chain novelty must hold, so the same contribution arriving from two sources is credited once; and the import boundary is an airgap, so a foreign chain's attestation is bonded and disputed, never trusted, and re-measured by the same value function rather than imported at its source's claimed worth. The merge conserves; it does not import another chain's trust assumptions.

14 Related work and what is novel

Three families of work are close to ours, and in each the difference is the same: what consensus is about is something other than a contribution's value measured on the chain itself.

Useful proof-of-work. Primecoin, Permacoin, proof-of-replication (Filecoin), proof of space-time (Chia), and the theoretical line of Ball, Rosen, Sabin and Vasudevan make the *work* useful, but value stays exogenous: security comes from puzzle hardness, dedicated space, or honest execution, while the work's worth is priced by an external market or fixed by an external problem-provider. That exogenous utility is exactly why these fall short as *value* chains: a prime, a protein fold, or a stored file is useful to someone *outside* the chain, never to the chain itself, so the chain still cannot read its own work to value or secure itself, and falls back on puzzle-hardness or an outside price. Proof of Mind inverts this. Its value is *endogenous*, the chain's own consensus object, so a contribution is genuinely useful *to the chain*: the chain measures it, prices it, and secures itself with it. Useful proof-of-work made the work mean something to the world; Proof of Mind makes it mean something to the system. The theoretical line is weaker than it appears (the usefulness claims of the 2017 construction were retracted in 2018), and proof-of-learning schemes are provably not robust to spoofing. Decentralized-training systems (Gensyn, Prime Intellect, Nous) make verified compute or training-integrity the consensus object, not output value. Decentralized-inference networks come closer, and we treat them next.

Output quality by peer agreement (Bittensor, Fortytwo). The closest operating systems do make output quality the reward object, and they are the strongest prior art. Bittensor's Yuma consensus takes a stake-weighted, clipped aggregate of validators' quality scores; Fortytwo aggregates pseudo-random pairwise comparisons among nodes with a Bradley–Terry model, the highest-quality response winning consensus and reputation tracking how often a node's outputs win. The differentiator is *how* the quality is obtained. Both rest on *subjective peer or validator agreement*: Yuma rewards agreeing with the stake-weighted majority, and empirically reward is dominated by purchased stake rather than by quality; Fortytwo's score is a peer ranking, not a value the protocol computes. Neither uses a cooperative-game synergy value over a provenance graph, neither binds the score to a learned value-model with a residual trust-certificate, and in Fortytwo reputation weights consensus without being the finality object itself. Noesis measures value objectively, by construction, and makes that measurement the finality object directly.

Contribution attribution. Ethereum's Deep Funding is the most superficially similar prior work, sharing the "value as a graph" framing and an AI-distilled credit signal. It is nonetheless distinct on two independent axes. It computes a distilled *scalar edge-weight* (the fraction of a dependent's credit assigned to a dependency), regressed by a model to match jury pairwise preferences; it is not a cooperative-game value, has no characteristic function, no coalition marginal contribution, and no connected-coalition restriction. And it is an off-chain *funding-allocation oracle* that splits a fixed pot, never a consensus, finality, or stake-weight object. Data-Shapley applies the flat (non-graph) Shapley value to training data as an offline metric; the Myerson value we use is the graph-restricted generalization (Myerson, 1977), a known primitive whose novelty here is its *application as the consensus object*, not the primitive itself. Every on-chain use of the Shapley value we found redistributes revenue *after* consensus rather than securing it.

Non-transferable standing. SourceCred separates non-transferable earned reputation (Cred, computed by PageRank) from a transferable token (Grain), and the Decentralized Society proposal introduces non-transferable soulbound tokens for reputation and credentials. The split pattern therefore has prior art. Neither, however, makes non-transferable earned standing the *consensus weight* while keeping a separate *transferable state-capacity* token: SourceCred's consensus is social, and the soulbound-token proposal addresses correlation-discounted DAO voting, not a consensus franchise.

Net. *Objective* value measurement as the consensus object, a cooperative-game value over a provenance graph made the finality weight directly (Section 4), is the novel core: the adjacent systems above reward output quality by subjective peer agreement, not by an objective measure. The Myerson-over-provenance-DAG credit rule and the soulbound-standing / transferable-capacity split are novel in their consensus application, though each composes known primitives. The HodgeRank residual as a stake-slashing manipulation alarm is *not* new: Delegated Proof of Reputation (2019) already proposes a Hodge-subspace filter that slashes flows in the harmonic and curl subspaces; our narrower contribution pairs the residual with a learned value-model and the endogenous value object, which that proposal lacks. We state these as positioned claims open to refutation, in the spirit of Section 10.

15 Value at the margin

Economics took a century to learn what this chain has to assume. Classical value theory located worth in cost: a thing was worth the labor or resources sunk into it. The diamond–water paradox broke that view (water sustains life and is nearly free; diamonds are useless and dear), and the marginal revolution of the 1870s — Jevons, Menger, and Walras, independently — replaced it. A thing is worth neither its total usefulness nor its cost of production but its value *at the margin*: the difference one more unit makes, revealed through exchange rather than fixed before it. Value is endogenous to the system of use, not a property carried in from outside.

Proof of work is a cost theory of value by design. It cannot read the worth of what a block contributes, so it prices security by energy expended — a cost — and leaves the coin's worth to an outside market. This is not a flaw to mock; it is the honest move for a chain with no way to measure marginal contribution. The complaint that the energy is "wasted" is the old cost-theory confusion run backward: cost was never worth, so spent cost cannot be waste measured against a value the work never claimed to set.

A value chain can measure marginal contribution, so it can price at the margin directly. The Myerson value of Section 4 is exactly that — a block's credit is its average marginal contribution across the orders in which the outcome could have been assembled, restricted to the coalitions its provenance actually connects — and realized downstream flow is the same idea read forward: a block is worth the difference it makes to what is later built, the marginal revolution's "one more unit" applied to contribution. Proof of Mind is marginal value theory made a consensus object: value endogenous, set at the margin, revealed by use, and now computed and secured on the very chain that produces it.

16 How honesty propagates upward

A base layer that makes honesty the profitable move does not keep that property to itself; anything built on top of it inherits it. On an ordinary chain every application has to re-fight the same battles on its own, front-running, price manipulation, value siphoned off in the middle, because the chain underneath does not pay for honesty. Each app patches the leak locally, and most of them lose. On a chain where the way to earn is to add real value that others build on, an application does not have to bolt honesty on as a feature. It is already the default underneath.

Take decentralized finance. Here, providing real liquidity or quoting an honest price is a contribution that earns, because someone downstream builds on it. Manipulating a market is the opposite: it moves value in a circle without adding anything anyone builds on, so it earns nothing and is slashed when challenged. "Honest DeFi" stops being a promise an application has to make and becomes the path of least resistance the chain already rewards. The same holds for any shared activity: because you are paid in proportion to what others genuinely build on your work, the smart move at every layer is to make your work worth building on. Cooperation is not requested; it is simply the highest-paying strategy — evolutionarily stable in the sense of Maynard Smith and Price — and it compounds, since each honest layer makes honesty cheaper for the layer above it. This is the same equilibrium seen one level up: a whole ecosystem in which the dishonest move is just the one that does not pay. This holds exactly to the degree the value measure is un-gameable — the load-bearing open problem of Section 10 — and is as strong as that measurement, no stronger.

17 Conclusion

We have specified a chain on which value is created as contribution, priced endogenously by a synergy-aware cooperative value over a provenance graph, owned and transferred like a UTXO, and used to secure consensus through Proof of Mind. The design replaces the inferences that make decentralized trust intractable, "is this a mind worth trusting," with proofs anyone can recompute, and it gives contribution the medium it never had, as Bitcoin gave possession one. The honest center of the system is a single hard problem, un-gameable value measurement, and the architecture is built so that the learned component charged with it can only ever deny value, never mint it, while a separate certificate reports when the value can be trusted at all. What remains is to feed that evaluator real outcomes at scale. The rest is assembled from known mathematics, and it runs.

Notation

Symbol	Meaning
b	a block: $\{\mathrm{id}, \mathrm{parent}, t, \mathrm{inputs}, \mathrm{output}, h\}$
$H(b \,\\|\, r),\ r$	commitment hash; reveal secret
$v(S)$	outcome value of a set S of blocks (the central object)
ϕ_i	Myerson value (synergy credit) of block / contributor i
$\nu(b)$	temporal novelty of b (coverage not claimed by an earlier block)
$\sigma,\ s_i$	logistic function; latent Bradley–Terry strength of i
r_θ	learned reward model (the production evaluator for v)
$\mathrm{seed}_i,\ \omega(\ell_i),\ \ell_i$	value-gate seed; learned outcome factor; provenance lineage of i
$\rho,\ \varphi$	saturation decay $\rho = 1/\varphi$; golden ratio $\varphi \approx 1.618$
$\mathrm{PoM}(a)$	Proof-of-Mind score of agent a (sum of ϕ over its owned blocks)
$W^d_{\mathrm{for}},\ W^d_{\mathrm{eff}},\ Q^d$	for- / effective- / quorum-floor vote weight in dimension d

(φ, bare, is the golden ratio in the saturation bound; ϕ_i, subscripted, is the Myerson credit. They are distinct.)

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